WORKED SOLUTION

STEP II 2010, Question 3

Pure • Fibonacci Numbers and Recurrences

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q3

The key idea

Four given terms and four unknowns: substitute $n = 0, 1, 2, 3$ into $(*)$ and solve. The trick is to divide one equation by another, which cancels a and turns the conditions into statements about $\lambda + \mu$ and $\lambda\mu$, the symmetric functions that make λ and μ the roots of a quadratic. Once the formula is known, the sum in (iii) splits into two geometric series.

Solution

Part (i)

Putting $n = 0, 1, 2, 3$ into $(*)$ gives

$$\begin{aligned} a + b &= 0, & a\lambda + b\mu &= 1, \\ a\lambda^2 + b\mu^2 &= 1, & a\lambda^3 + b\mu^3 &= 2. \end{aligned}$$

The first gives $b = -a$, so the other three become

$$a(\lambda - \mu) = 1, \qquad a(\lambda^2 - \mu^2) = 1, \qquad a(\lambda^3 - \mu^3) = 2.$$

In particular $a(\lambda - \mu) = 1$ tells us that $a \neq 0$ and $\lambda \neq \mu$. Factorising,

$$\lambda^2 - \mu^2 = (\lambda - \mu)(\lambda + \mu), \qquad \lambda^3 - \mu^3 = (\lambda - \mu)(\lambda^2 + \lambda\mu + \mu^2),$$

so dividing the second and third equations by $a(\lambda - \mu) = 1$ (allowed, as it is non-zero) gives

$$\lambda + \mu = 1 \qquad \text{and} \qquad \lambda^2 + \lambda\mu + \mu^2 = 2,$$

the second of which is the required result.

? **Ask yourself** How do we get λ and μ themselves from these two equations?

Write the second in terms of the sum and product: $\lambda^2+\lambda\mu+\mu^2=(\lambda+\mu)^2-\lambda\mu$, so $1-\lambda\mu=2$ and $\lambda\mu=-1$. Numbers with sum 1 and product -1 are the roots of

$$t^2-t-1=0, \qquad \text{so} \qquad t=\frac{1\pm\sqrt{5}}{2}.$$

Which root is λ? From $a(\lambda-\mu)=1$ and $a>0$ we need $\lambda-\mu>0$, so λ is the larger root. Then $\lambda-\mu=\sqrt{5}$, and

$$\lambda=\frac{1+\sqrt{5}}{2}, \qquad \mu=\frac{1-\sqrt{5}}{2}, \qquad a=\frac{1}{\sqrt{5}}, \qquad b=-\frac{1}{\sqrt{5}}.$$

(The other labelling would force $a=-1/\sqrt{5}$, which is ruled out.) Conversely, these values satisfy all four equations, since every step above can be reversed.

Part (ii)

We have $F_6=(\lambda^6-\mu^6)/\sqrt{5}$. Rather than expanding $(\frac{1+\sqrt{5}}{2})^6$ directly, use the fact that λ is a root of $t^2=t+1$ to reduce powers:

$$\begin{aligned}
\lambda^2&=\lambda+1,\\
\lambda^3&=\lambda^2+\lambda=2\lambda+1,\\
\lambda^6&=(2\lambda+1)^2=4\lambda^2+4\lambda+1=4(\lambda+1)+4\lambda+1=8\lambda+5.
\end{aligned}$$

Exactly the same holds for μ, which satisfies the same quadratic, so $\mu^6=8\mu+5$. Hence

$$F_6=\frac{(8\lambda+5)-(8\mu+5)}{\sqrt{5}}=\frac{8(\lambda-\mu)}{\sqrt{5}}=\frac{8\sqrt{5}}{\sqrt{5}}=8.$$

Part (iii)

Using $(*)$,

$$\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}=\frac{1}{2\sqrt{5}}\sum_{n=0}^{\infty}\left[\left(\frac{\lambda}{2}\right)^n-\left(\frac{\mu}{2}\right)^n\right].$$

Since $\lambda\approx 1.618$ and $\mu\approx -0.618$, both common ratios $\lambda/2$ and $\mu/2$ have modulus less than 1. So both geometric series converge, and we may split the sum into the difference of the two:

$$\begin{aligned}
\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}&=\frac{1}{2\sqrt{5}}\left[\frac{1}{1-\lambda/2}-\frac{1}{1-\mu/2}\right]=\frac{1}{\sqrt{5}}\left[\frac{1}{2-\lambda}-\frac{1}{2-\mu}\right]\\
&=\frac{1}{\sqrt{5}}\cdot\frac{\lambda-\mu}{(2-\lambda)(2-\mu)}=\frac{1}{4-2(\lambda+\mu)+\lambda\mu},
\end{aligned}$$

using $\lambda-\mu=\sqrt{5}$. With $\lambda+\mu=1$ and $\lambda\mu=-1$ the denominator is $4-2-1=1$, so

$$\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}=1.$$

Checks

The formula should give integers: $F_4 = (\lambda^4 - \mu^4)/\sqrt{5}$ with $\lambda^4 = 3\lambda + 2$ gives $F_4 = 3$, and $F_5 = 5$, $F_6 = 8$ fit the pattern $F_{n+2} = F_{n+1} + F_n$ seen in the first four terms. For (iii), the partial sums 0, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{19}{32}$, $\ldots$ creep up towards 1; adding the first 300 terms numerically gives 1.0000000 to machine precision. The terms decay only like $(0.809)^n$, so convergence is slow.

Common slips

- Forgetting to use $a > 0$ to decide which root is λ: without it the answer isn't determined.
- Dividing by $\lambda - \mu$ without noting why it is non-zero (it is, because $a(\lambda - \mu) = 1$).
- Splitting the infinite sum into two series without checking that each converges.

Beyond STEP

The formula here is Binet's formula for the Fibonacci numbers, and the method is the general one for linear recurrences: λ and μ are the roots of the characteristic equation $t^2 = t + 1$ of $F_{n+2} = F_{n+1} + F_n$. Part (iii) evaluates the generating function $\sum F_n x^n = x/(1 - x - x^2)$ at $x = \frac{1}{2}$ (and halves it), which gives 1 again. Generating functions are a major tool in first-year combinatorics and probability; for the golden ratio λ and the Fibonacci numbers in the theory of continued fractions, see Davenport, *The Higher Arithmetic*.

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